

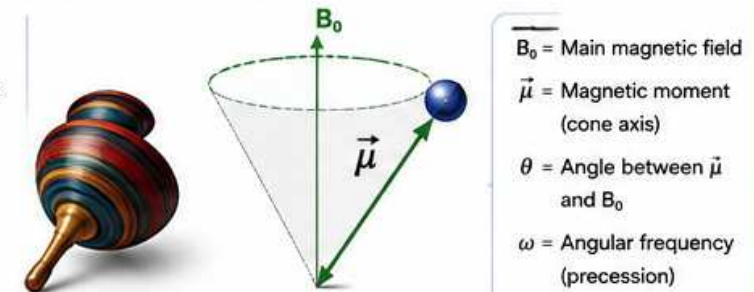
1.4 PRECESSION – SPINNING LIKE A TOP

*Aligned protons do not point straight up.
They precess around the B_0 field like a spinning top.*

Even though more protons align with the main magnetic field (B_0), their magnetic moments are not fixed in place. Each proton “wobbles” around the B_0 direction while pointing slightly upward. This motion is called **PRECESSION**.

THE CONCEPT

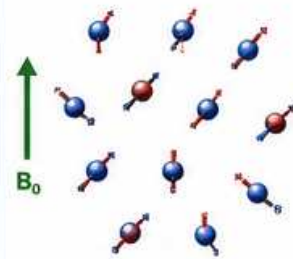
A spinning top stays upright while its axis precesses around gravity. Similarly, a proton's magnetic moment (μ) precesses around the main magnetic field (B_0) while maintaining a constant angle.



WHAT HAPPENS TO AN ALIGNED PROTON?

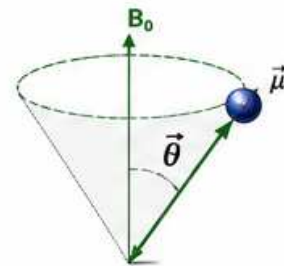
1 Apply B_0 Field

Protons start to align with B_0 .



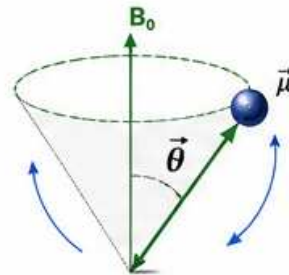
2 Start of Precession

Proton's magnetic moment is tipped and begins to wobble (precess).



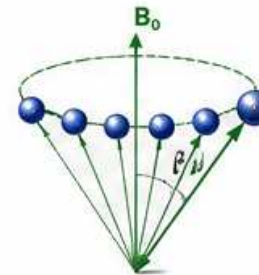
3 Continuous Precession

μ rotates around B_0 at a constant angle θ .



4 Many Protons Together

All aligned protons precess in phase (at thermal equilibrium).



Precession is the fundamental motion that allows MRI to use RF energy to excite and detect signals.

KEY EQUATIONS

Angular precession frequency (ω)

$$\omega = \frac{d\phi}{dt} = \gamma B_0 \quad (\text{rad/s})$$

Larmor frequency (f)

$$f = \frac{\omega}{2\pi} = \frac{\gamma}{2\pi} B_0 \quad (\text{Hz})$$

Gyromagnetic ratio for hydrogen (γ)

$$\gamma/2\pi = 42.58 \text{ MHz/T}$$

EXAMPLE (HYDROGEN)

At 1.5 Tesla:

$$f = 42.58 \text{ MHz/T} \times 1.5 \text{ T} = 63.87 \text{ MHz}$$

At 3.0 Tesla:

$$f = 42.58 \text{ MHz/T} \times 3.0 \text{ T} = 127.74 \text{ MHz}$$

At 7.0 Tesla:

$$f = 42.58 \text{ MHz/T} \times 7.0 \text{ T} = 298.06 \text{ MHz}$$

ANALOGY: SPINNING TOP



Axis of the top (precession axis)
Path of precession (circular motion)
Top body (magnetic moment)

KEY POINTS

- ✓ Precession occurs around the B_0 field.
- ✓ The angle (θ) is small at thermal equilibrium (only a few degrees).
- ✓ The faster the field (B_0), the faster the precession.
- ✓ Precession frequency is the same for all protons in the same field.

FIELD SERVICE TIP

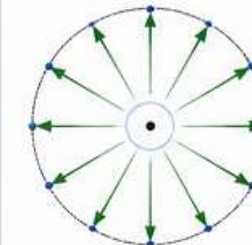
- B_0 field uniformity is critical.
If B_0 is uneven:
- Precession frequency varies across space.
 - Protons go out of sync (dephase).
 - Signal is lost (broad lines, poor image).

Always ensure proper shimming!

COMMON MISCONCEPTION

Protons do NOT spin around their own axis like a ball.
They precess around B_0 , just like the axis of a spinning top precesses around gravity.

VISUALIZING PRECESSION



Top view looking along B_0 direction.
The tip of μ traces a circle (cone top) while precessing.

KEY TAKEAWAY

Precession is the “wobble” of proton magnetic moments around the main magnetic field (B_0). It occurs at the Larmor frequency and is the foundation for RF excitation and MRI signal detection.

WHAT'S NEXT? >>

Next Section 1.5 – Larmor Frequency
We'll see how precession speed depends on B_0 and leads to the Larmor frequency.